



Modelling Financial Markets

Chapter 13: Machine Learning in Finance



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/3)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AUC: Area Under the (ROC) Curve
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CPCV: Combinatorial Purged Cross-Validation
- ▣ CRSP: Center for Research in Security Prices
- ▣ CSCV: Combinatorially Symmetric Cross-Validation
- ▣ CV: Cross-Validation
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ DML: Double/Debiased Machine Learning
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ELW: Exact Local Whittle (estimator)
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ EV: Explained Variation
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FFD: Fixed-width window Fractional Differentiation

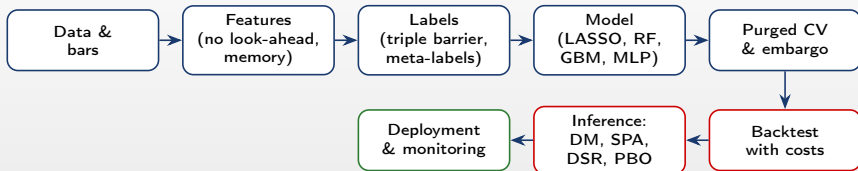
Acronyms used in this material (2/3)

- ▣ **FFN: FeedForward Network**
- ▣ **FN: False Negative**
- ▣ **FRED-MD: Federal Reserve Economic Data, Monthly Database (McCracken–Ng)**
- ▣ **FWER: Family-Wise Error Rate**
- ▣ **GAN: Generative Adversarial Network**
- ▣ **GBM: Gradient Boosting Machine**
- ▣ **GMM: Generalized Method of Moments**
- ▣ **GRS: Gibbons–Ross–Shanken (test)**
- ▣ **GRU: Gated Recurrent Unit**
- ▣ **GW: Giacomini–White (test)**
- ▣ **HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)**
- ▣ **HML: High Minus Low (value factor)**
- ▣ **IBM: International Business Machines (Corporation)**
- ▣ **IS: In-Sample**
- ▣ **LASSO: Least Absolute Shrinkage and Selection Operator**
- ▣ **LIME: Local Interpretable Model-agnostic Explanations**
- ▣ **LLM: Large Language Model**
- ▣ **LS: Linear SDF estimated by unregularised Least Squares**
- ▣ **LSTM: Long Short-Term Memory (network)**
- ▣ **LW: Local Whittle (estimator)**
- ▣ **MA: Moving Average**
- ▣ **MA20: 20-day Moving Average**
- ▣ **MDA: Mean Decrease Accuracy**
- ▣ **MDI: Mean Decrease Impurity**
- ▣ **ML: Machine Learning**
- ▣ **MLP: MultiLayer Perceptron**

Acronyms used in this material (3/3)

- MSCA: Marie Skłodowska-Curie Actions
- OLS: Ordinary Least Squares
- OOS: Out-Of-Sample
- PAC: Probably Approximately Correct (learning)
- PBO: Probability of Backtest Overfitting
- PCR: Principal Component Regression
- PLS: Partial Least Squares
- PSR: Probabilistic Sharpe Ratio
- PT: Pesaran–Timmermann (test of directional accuracy)
- RC: Reality Check (White's test for data snooping)
- RF: Random Forest
- RL: Reinforcement Learning
- RMW: Robust Minus Weak (profitability factor)
- ROC: Receiver Operating Characteristic (curve)
- RSI: Relative Strength Index
- SD: Standard Deviation
- SDF: Stochastic Discount Factor
- SE: Standard Error
- SHAP: SHapley Additive exPlanations
- SPA: Superior Predictive Ability (test)
- SR: Sharpe Ratio
- TN: True Negative
- TP: True Positive
- US: United States
- VIX: CBOE Volatility IndeX
- WF: Walk-Forward (validation)

Chapter Route: the Financial ML Workflow



- ▣ Part I: the prediction problem, a minimal toolbox, features and fractional differencing, labels, honest validation
- ▣ Part II: an S&P 500 case study with formal tests, selection and backtest overfitting, machine learning (ML) for inference, a research case study, AI for scientific discovery; technical detail sits in the Appendix
- ▣ After this chapter you can: build features and labels without look-ahead; validate with purging and embargo; test a classifier and a backtest against a benchmark (PT, DM, DSR, SPA, PBO)
- ▣ **Look-ahead bias**: using information not available at decision time; **selection bias**: reporting the best of many trials; [López de Prado \(2018\)](#): most failures come from the process, not the algorithm

Why Machine Learning in Finance, and Why It Is Hard

The promise

- Flexible, non-linear functional forms; interactions found automatically
- Hundreds of candidate predictors (the “factor zoo”, Chapter 3)
- Alternative data: text, images, order books, on-chain data

The constraints

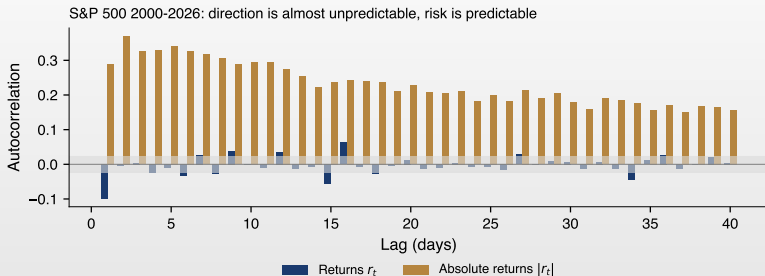
- Very low **signal-to-noise ratio**; one serially dependent history that cannot be re-run
- **Non-stationarity**: regimes, breaks; markets react to their users
- Labels must be constructed; errors cost real money
- Massive risk of **overfitting** (fitting noise)

Adaptive Markets Hypothesis: Lo (2004)

- Predictability appears, is arbitrated away and re-appears: a model that worked yesterday is a **hypothesis**, not a law (market efficiency: Chapter 2)
- The hard part is not fitting a model but **not fooling yourself**: correct labels, correct validation, honest backtests



What Is Predictable? Direction vs. Risk



- Autocorrelation at lag k : $\rho_k = \text{Corr}(x_t, x_{t-k})$; the grey band is the 95% no-correlation band $\pm 1.96/\sqrt{T}$
- Autocorrelations of returns r_t are tiny (inside the band); those of $|r_t|$ are large and persistent
- ML has the best chance where there is structure: **volatility, risk, liquidity, cross-sectional ranking**

Evidence, and the Economic Value of a Small R_{OOS}^2

- [Gu, Kelly & Xiu \(2020\)](#): monthly returns of individual US stocks; linear models, penalised regressions, principal component regression (PCR), partial least squares (PLS), forests, boosted trees, neural nets
 - ▶ $R_{OOS}^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$ is small (well below 1% a month); trees and nets win through **non-linear interactions**; momentum, liquidity and volatility dominate
- [Campbell & Thompson \(2008\)](#): in $r_{t+1} = \mu + x_t + \varepsilon_{t+1}$ with $R^2 = \text{Var}(x_t) / \text{Var}(r_{t+1})$, an unconstrained mean-variance investor who knows x_t (no costs, no estimation error) reaches

$$SR^{*2} = \frac{SR_0^2 + R^2}{1 - R^2}, \quad SR_0 = \text{Sharpe ratio of buy-and-hold}$$

- ▶ S&P 500, monthly log returns 2001–2026: $SR_0 = 0.128$ a month (0.44 annualised); $R^2 = 0.5\%$ a month $\Rightarrow SR^* = 0.51$ (+15%), $R^2 = 1\% \Rightarrow 0.57$ (+27%)
- The benchmark matters: [Welch & Goyal \(2008\)](#) show that most published predictors lose to the **historical mean**; [Gu, Kelly & Xiu \(2020\)](#) and Chapter 14 use a zero forecast
- For a **single index** the effective sample is far smaller than for thousands of stocks: the S&P 500 case study tests exactly this

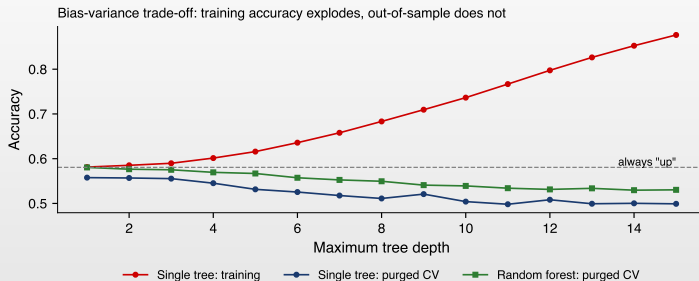
Loss, Penalty and Five Model Families

- Supervised learning: $\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_i L(y_i, f(x_i)) + \lambda \Omega(f)$; loss L (squared error, log-loss), complexity penalty Ω , tuning $\lambda \geq 0$ chosen by K -fold cross-validation (CV)
 - expected error = bias² + variance + noise σ^2 (Hastie, Tibshirani & Friedman (2009), Ch. 7); in finance σ^2 dominates, so variance control matters more than flexibility
- Penalised logit / least squares: $\lambda [\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2]$: Ridge $\alpha = 0$ (Hoerl & Kennard (1970)), LASSO $\alpha = 1$ (Tibshirani (1996)), Elastic Net in between (Zou & Hastie (2005)); the linear benchmark is the logit $p = 1/(1 + e^{-x^\top \beta})$
- Trees: recursive splits minimising Gini impurity; random forest (RF): average of B decorrelated bootstrap trees (Breiman (2001)); its out-of-bag error is **optimistic** when labels overlap
- Gradient boosting machine (GBM): $F_m = F_{m-1} + \eta h_m$, small trees fitted to the negative gradient (Friedman (2001))
- Multilayer perceptron (MLP): $h^{(\ell)} = g(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$, ReLU g , early stopping; architectures in Chapter 14
- This chapter classifies the 5-day direction; every CV score is an estimate and is tested against a benchmark



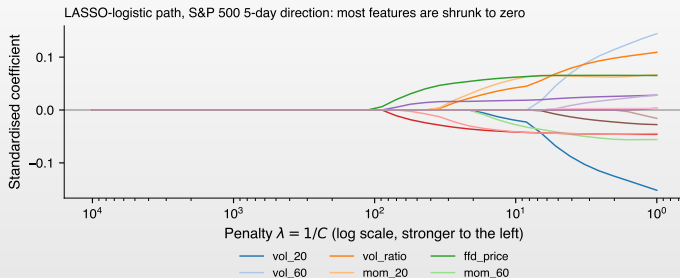
Random Forests

Bias–Variance in Practice: Tree Depth



- S&P 500, 5-day direction; accuracy = share of correctly classified days
- Training accuracy rises to $\approx 87\%$ with depth; purged-CV accuracy **falls** towards 50%
- Random forests degrade more slowly, but never beat the “always up” baseline (dashed line)

LASSO Regularisation Path



- L_1 -penalised logistic regression for the 5-day direction of the S&P 500, 14 standardised features
- Strong penalty: every coefficient is zero; the first feature to enter is the FFD price, a scaled price level (next section)
- Weak penalty: the volatility features (vol_20, vol_60, vol_ratio) get the largest coefficients
- All coefficients are small in absolute value: weak signal, consistent with near-efficient markets

Stationarity, Memory and the Fractional Difference

- ML inputs need a distribution that does not drift; log prices behave as $I(1)$, returns $\Delta \log P_t$ are $I(0)$ but **erase the memory** of the price level
- A series is $I(\delta)$ if $(1 - B)^\delta X_t$ is $I(0)$, with backshift $BX_t = X_{t-1}$; it is stationary only for $\delta < \frac{1}{2}$ (Hosking (1981))

Definition 1 (Fractional difference of order d)

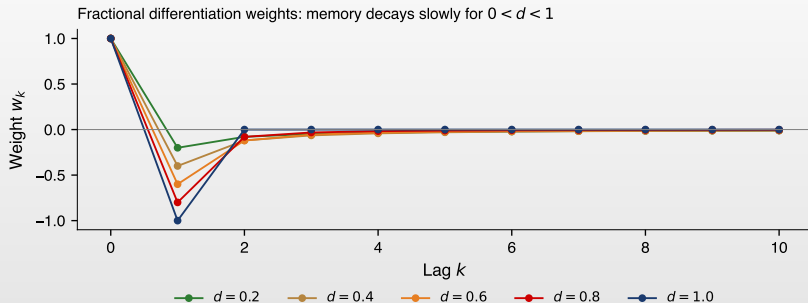
- $\tilde{X}_t = (1 - B)^d X_t = \sum_{k=0}^{\infty} w_k X_{t-k}$, with $w_0 = 1$ and $w_k = -w_{k-1} \frac{d-k+1}{k}$
- $d = 1$: $w = (1, -1, 0, \dots)$, ordinary returns; $0 < d < 1$: weights decay hyperbolically, so memory is kept
- An $I(1)$ log price differenced with d is $I(1 - d)$: non-stationary for $d \leq \frac{1}{2}$, stationary with long memory for $\frac{1}{2} < d < 1$
- The infinite weights sum to $(1 - 1)^d = 0$: the level cancels only in the limit (asymptotics in the Appendix)
- López de Prado (2018) proposes the minimum d at which an ADF test rejects; this section shows why that rule fails

The Fixed-Width Window (FFD)

Definition 2 (Fixed-width-window fractional differentiation (FFD))

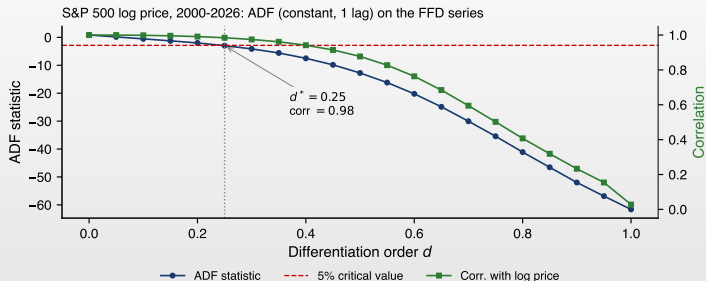
- ▣ Truncate the weights: keep $w_0, \dots, w_K$ with $|w_k| \geq \tau$ (e.g. $\tau = 10^{-4}$), drop the rest (López de Prado (2018))
- ▣ $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$ for every $t > K$
- ▣ Every $\tilde{X}_t$ uses the same number of lags; only past values enter $\Rightarrow$ no look-ahead
- ▣ The smaller d , the more lags are needed: $d = 0.25, \tau = 10^{-4} \Rightarrow K = 444$ lags
- ▣ The truncated weights no longer sum to zero: $\sum_{k=0}^K w_k = 0.178$ at $d = 0.25$ (0.040 at $d = 0.5$)

Fractional Differentiation Weights



- For $d = 1$ only the first lag matters; for small d the weights of distant lags remain non-zero
- All weights after w_0 are negative: $\tilde{X}_t$ compares today's level with a weighted history

The ADF Statistic Along d : S&P 500



- Augmented Dickey–Fuller (ADF) unit-root test (Time Series course): H_0 unit root, reject below ≈ -2.86 at 5%
- FFD log price, S&P 500 2000–2026, constant and 1 lag: the ADF rejects from $d = 0.25$, where $\text{Corr}(\tilde{X}_t, \log P_t) = 0.98$
- Same series, lag length chosen by AIC (34 lags): ADF = -0.72 , no rejection
- A correlation of 0.98 with an $I(1)$ price is a warning, not a virtue

Truncated FFD of an $I(1)$ Price Is Still $I(1)$

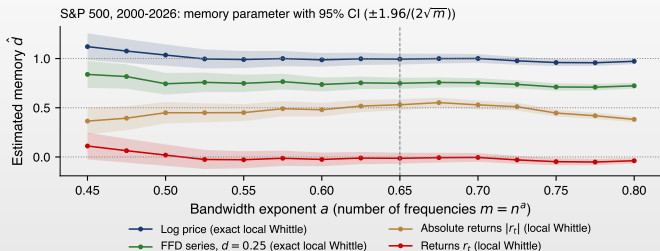
- Decompose the FFD series:

$$\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k} = \left(\sum_{k=0}^K w_k \right) X_t - \sum_{k=1}^K w_k (X_t - X_{t-k})$$

- If X_t is $I(1)$, each $X_t - X_{t-k}$ is $I(0)$; the first term is $0.178 X_t$ at $d = 0.25$: a **scaled random walk**
- The unit-root component has small variance next to the stationary part, so ADF with few lags confuses the two
 - Monte Carlo: pure random walk, $T = 6,274$, FFD with $d = 0.25$: ADF (1 lag) rejects in **97.8%** of 500 samples, at a nominal 5%
- Consequence: `ffd_price` in the case study is a scaled price level, not a stationary feature; the next slide measures the memory directly

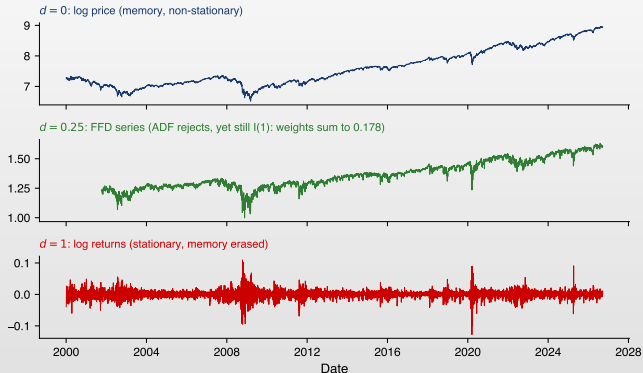
 `inference_tests.py`

Memory Estimated Directly: Local Whittle across Bandwidths



- An $I(\delta)$ series has spectral density $f(\lambda) \sim G\lambda^{-2\delta}$ near 0; local Whittle (LW, Robinson (1995)) fits it on the first $m = n^\alpha$ Fourier frequencies, exact local Whittle (ELW, Shimotsu & Phillips (2005)) also for $\delta \geq \frac{1}{2}$; 95% CI $\hat{\delta} \pm 1.96/(2\sqrt{m})$
- S&P 500, 2000–2026, $m = n^{0.65}$: log price (ELW) 0.99 [0.94; 1.05]; FFD series (ELW) 0.75 [0.69; 0.81]; returns -0.01 $[-0.07; 0.04]$; $|r_t|$ 0.53 [0.47; 0.59]
- The FFD series stays near $1 - d$ over the band (its order is still 1); the long memory is in **volatility** (Chapters 5 and 9): fractional features belong on $|r_t|$ (Franke, Härdle & Hafner (2019))

Prices, FFD Series and Returns



- ▣ **Top:** log price — $I(1)$
- ▣ **Middle:** FFD with $d = 0.25$ — the upward trend of the price is still visible: $0.178 \log P_t$ plus a stationary part
- ▣ **Bottom:** log returns — only volatility clusters are visible
- ▣ The FFD series enters the case study as feature `ffd_price`

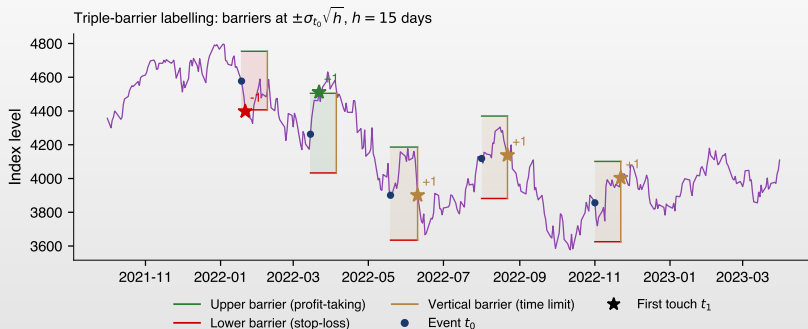
From Fixed-Horizon to Triple-Barrier Labels

- ▣ **Fixed horizon:** $r_{t,t+h} = \log P_{t+h} - \log P_t$, threshold $\tau \geq 0$: $y_t = +1$ if $r_{t,t+h} > \tau$, -1 if $r_{t,t+h} < -\tau$, 0 otherwise
 - ▶ a constant τ ignores time-varying volatility, the path ignores stop-losses, and the classes are unbalanced: S&P 500, $h = 10$, $\tau = 0$: 60.5% up vs. 39.5% down

Definition 3 (Triple-barrier label, López de Prado (2018))

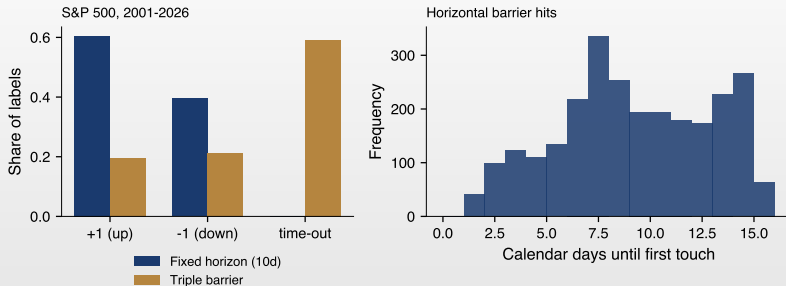
- ▣ At event time t_0 , with daily volatility σ_{t_0} and horizon h : upper barrier $\log P_t - \log P_{t_0} \geq \pi_u \sigma_{t_0} \sqrt{h}$ (profit-taking), lower barrier $\leq -\pi_l \sigma_{t_0} \sqrt{h}$ (stop-loss), vertical barrier $t = t_0 + h$
- ▣ Label = **first barrier touched** at t_1 : $+1$ upper, -1 lower; at the vertical barrier $\text{sign}(r_{t_0,t_1})$ (or 0)
- ▣ σ_{t_0} : square root of the exponentially weighted moving average (EWMA) of squared deviations of daily log returns, span 50 ($\lambda = 1 - 2/51$): barriers scale with volatility, like a trader's take-profit, stop-loss and maximum holding period
- ▣ Each label has a **span** $[t_0, t_1]$: the input of purging in validation

Triple Barrier: S&P 500 in 2022



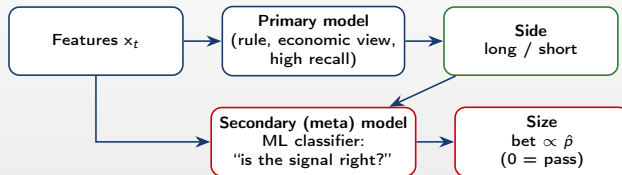
- Five events during the 2022 bear market, $\pi_u = \pi_l = 1$, $h = 15$ days
- January: stop-loss hit (-1); March: profit-taking ($+1$); the others end at the vertical barrier

Label Distribution: Fixed Horizon vs. Triple Barrier



- S&P 500, 2001–2026, $h = 10$: fixed horizon gives 60.5% up / 39.5% down
- Triple barrier ($\pm 1\sigma\sqrt{h}$): 19.6% up, 21.3% down, 59.0% time-out — balanced directional labels
- Right panel: horizontal barriers are typically hit after 1–2 weeks

Meta-Labeling: Deciding Whether to Act



- Meta-label $y_t^{meta} = 1$ if the primary signal was profitable (barrier touched in its direction), 0 otherwise; the secondary model gives $\hat{p} = P(y_t^{meta} = 1 \mid x_t)$
- ML learns **when to trust** an interpretable primary rule, not the direction itself: precision can rise at the cost of recall
- Seminar B9, Bitcoin MA20/MA50 crossover, 1,660 test trades: precision $0.475 \rightarrow 0.517$, recall $1 \rightarrow 0.541$, F1 $0.644 \rightarrow 0.529$; gross return per trade $-0.27\% \rightarrow +0.19\%$
- Filtering need not improve F1; with costs per trade, precision and the return per trade are what matter

Overlapping Labels and Sample Uniqueness

- Label i spans $[t_{0,i}, t_{1,i}]$; consecutive labels share returns $\Rightarrow$ samples are **not i.i.d.**
- Concurrent labels at t : $c_t = \sum_i 1\{t_{0,i} \leq t \leq t_{1,i}\}$; average uniqueness $\bar{u}_i = \frac{1}{t_{1,i} - t_{0,i} + 1} \sum_{t=t_{0,i}}^{t_{1,i}} \frac{1}{c_t}$
- $\bar{u}_i = 1$: no shared return; $\bar{u}_i = 1/5$: the harmonic mean of c_t over the span of label i is five

Consequences

- Effective sample size $\approx \sum_i \bar{u}_i \ll n$ (an overlap heuristic, not a variance-based measure)
- Bagging draws near-duplicates; CV scores too optimistic

Remedies

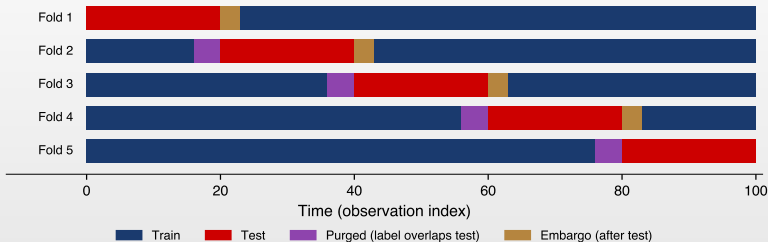
- Weight samples by $\bar{u}_i$ (and by $|r|$)
- Sequential bootstrap; in RF `max_samples` $\approx \bar{u}$

Leakage: What Crosses the Train/Test Boundary

Definition 4 (Leakage)

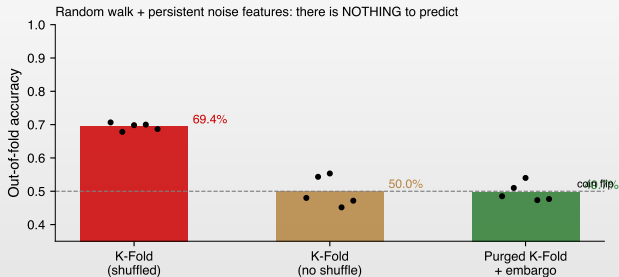
- The training set contains information that is also present in the test set, so the test score no longer measures performance on genuinely new data
- Serially correlated features ($x_t \approx x_{t+1}$) and overlapping labels ($y_t \approx y_{t+1}$): a shuffled K -fold puts **near-identical** days on both sides
- A flexible model (RF, GBM) memorises the neighbour $\Rightarrow$ excellent “out-of-sample” score; tuning on a leaky CV selects the **most overfitted** model
- Other sources: transformations fitted on the full sample (scaling, principal component analysis, the choice of d for FFD), survivorship bias, data revisions, wrong timestamps

Purged K-Fold with Embargo



- Folds are **contiguous** blocks in time (López de Prado (2018)); **purging** removes every training observation whose label span $[t_{0,i}, t_{1,i}]$ overlaps the span of the test labels, from the first test t_0 to the last test t_1
- **Embargo**: also remove a small share (e.g. 1%) of observations right after the test block, because features look back in time
- One boundary: labels end 2 days after they start, test $\{4, 5\}$: observations 2 and 3 end inside the test block (purged), 6 is the embargo, 7 starts before the last test label ends (purged); full table in the Appendix

Leakage Experiment: Nothing to Predict



- Price = pure random walk; label = sign of the 20-day forward return; 5 features = 20-day rolling means of independent noise
- True accuracy is 50% by construction; random forest, 5 folds
- **Shuffled K-fold: 69.4%**; contiguous K-fold 50.0%; purged K-fold + embargo 49.7%
- Shuffling fails because labels overlap *and* features persist; a result that looks too good is usually leakage

 MFM_ch13_purged_cv

Transformations on Training Data Only: the FFD Ablation

- The “minimum d ” was chosen on 2000–2026 and then used inside purged CV: the test folds helped choose a feature
 - ▶ the same ADF rule on data up to 2009 gives $d = 0.15$; on the full sample $d = 0.25$
- Case study without `ffd_price` (purged 5-fold, mean over folds)
 - ▶ accuracy: logit 57.5%, RF 56.4%, GBM 55.6%, MLP 55.5%, all below 58.1%
 - ▶ AUC: 0.483, 0.522, 0.514, 0.514
- The negative conclusion does not depend on this feature
- Correct protocol
 - ▶ estimate d (local Whittle, with its CI) inside each training fold only
 - ▶ prefer features with an economic meaning: distances to moving averages, fractional filters of $|r_t|$

 `inference_tests.py`

Model Selection versus Deployment

- Ordinary K -fold is not always invalid: for a purely autoregressive model with serially uncorrelated errors it is valid (Bergmeir, Hyndman & Koo (2018))
 - ▶ it fails when the errors are serially correlated: overlapping labels, persistent features outside the model's lags; check the autocorrelation of the CV residuals first
- **Model selection:** purged K -fold, or combinatorial purged CV (CPCV), which tests on many combinations of groups and returns a **distribution** of Sharpe ratios instead of one number
- **Deployment:** walk-forward (WF), train on $[0, T_k]$, test on $(T_k, T_{k+1}]$, roll forward; realistic, but a single path with high variance
- Rule of thumb: purged K -fold or CPCV to choose, walk-forward for the final, realistic evaluation (details in the Appendix)

Case Study: Data, Target and Validation

- ▣ S&P 500 index and VIX, daily closes, 2001–2026: $n = 6,269$ observations
- ▣ Target: $y_t = 1\{\log P_{t+5} - \log P_t > 0\}$, the 5-day direction; label span $[t, t + 5]$ used for purging
- ▣ Validation: purged 5-fold CV, 1% embargo; every model gets the same folds
- ▣ Metrics: accuracy and receiver operating characteristic (ROC) area under the curve (AUC), against the “always up” baseline
 - ▶ **fold mean**: the metric computed in each of the 5 test folds, then averaged
 - ▶ **pooled**: the metric computed once on all out-of-fold predictions together; used by the tests
- ▣ Chapter data: `data/market/GSPC.INDX.csv` and `VIX.INDX.csv`

Case Study: 14 Features and Four Models

Features, information up to day t

- ▣ **Momentum:** $\log P_t - \log P_{t-h}$, $h \in \{1, 5, 20, 60, 120\}$
- ▣ **Realised volatility:** annualised SD of daily log returns over 20 and 60 days, and their ratio
- ▣ **Distance to moving average:** $\log P_t$ minus the mean of $\log P$ over 50 or 200 days
- ▣ **RSI(14):** $100 - 100/(1 + RS)$, RS = smoothed average gain over average loss, 14 days
- ▣ **VIX:** log level and 5-day change
- ▣ **FFD** log price, $d = 0.25$, chosen on the full sample (ablation in the validation section)

Models, deliberately conservative

- ▣ **Logit:** L2-penalised (Ridge), standardised features
- ▣ **RF:** 400 trees, minimum leaf 100
- ▣ **GBM:** histogram-based, depth 3, learning rate 0.03, 200 trees
- ▣ **MLP:** hidden layers of 32 and 16 units, early stopping
- ▣ Large leaves and shallow trees limit variance

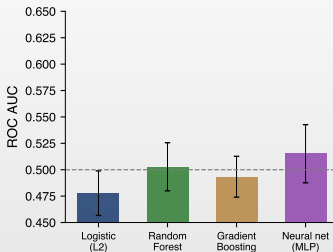
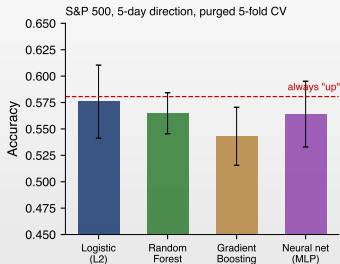
The Model That Learned “Always Up”

	Pred. up	Pred. down
Actual up	TP = 3,572	FN = 68
Actual down	FP = 2,591	TN = 38

- L2-penalised logit, pooled out-of-fold predictions of purged 5-fold CV, $n = 6,269$
- TP, FN, FP, TN: true positives, false negatives, false positives, true negatives

- Accuracy = $(TP + TN)/n = 57.6\%$; precision = $TP/(TP + FP) = 0.580$; recall = $TP/(TP + FN) = 0.981$; $F_1 = 0.729$
- The model predicts “up” on 98.3% of days: it has essentially learned “always up”
- **Majority baseline** from the same table: actual up days $(3,572 + 68)/6,269 = 3,640/6,269 = 58.1\%$
- A 57.6%-accurate model is *worse* than doing nothing
- Other baselines: yesterday’s direction, a moving-average rule, buy-and-hold for trading
- A model has value only if it beats the baseline **out of sample**, after costs, by more than sampling error

Model Comparison, Fold Means



Fold mean	Logit	RF	GBM	MLP
Accuracy	57.6%	56.5%	54.3%	56.4%
AUC	0.478	0.503	0.493	0.515

- Bars: mean over the 5 purged folds; whiskers: ± 1 SD across folds
- Baseline “always up”: **58.1%**; no model beats it

Are the Differences Significant? Pesaran–Timmermann and AUC Intervals

- **Pesaran–Timmermann** (PT, [Pesaran & Timmermann \(1992\)](#)): H_0 : predicted sign independent of realised sign; $\hat{P}$ = share correct, $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$, $PT = (\hat{P} - \hat{P}^*) / \widehat{SE} \rightarrow N(0, 1)$ (Appendix); with overlapping labels, the slope t of $y_t = a + b x_t + u_t$ with Newey–West errors ([Newey & West \(1987\)](#))
- **AUC**: the DeLong SE assumes independence; a circular block bootstrap ($L = 20$ days, [Künsch \(1989\)](#)) keeps the serial dependence

Pooled out-of-fold	Logit	RF	GBM	MLP
$\hat{P} / \hat{P}^*$	0.576 / 0.578	0.565 / 0.570	0.543 / 0.557	0.564 / 0.565
$PT / HAC\ t\ (\text{lag } 4)$	-1.28 / -1.00	-1.61 / -1.16	-3.07 / -2.31	-0.36 / -0.26
AUC	0.479	0.495	0.490	0.512
block-bootstrap 95% CI	[0.455; 0.504]	[0.472; 0.520]	[0.468; 0.514]	[0.489; 0.537]

- No significant positive directional skill; every AUC interval contains 0.5; GBM is significantly *worse* than independence
- Consistent with weak-form efficiency (Chapter 2); look at cross-sections ([Gu, Kelly & Xiu \(2020\)](#)), risk targets, crypto delisting ([Będowska-Sójka, Wójcik & Pele \(2026\)](#)), text (Chapter 15)

Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White

- ▣ Loss differential $d_t = L(y_t, \hat{p}_{1,t}) - L(y_t, \hat{p}_{0,t})$, log-loss or Brier loss $(y_t - \hat{p}_t)^2$; $d_t > 0$: model 1 worse
- ▣ **Diebold–Mariano** (DM, Diebold & Mariano (1995)): $DM = \bar{d} / \sqrt{\hat{\Omega}_{NW}/n} \rightarrow N(0, 1)$ under $H_0 : \mathbb{E}[d_t] = 0$; $\hat{\Omega}_{NW}$ = heteroskedasticity and autocorrelation consistent (HAC) long-run variance
- ▣ **Nested models** (the logit with 14 features nests the constant, $\beta = 0$): under H_0 the large model estimates zeros with noise, $\mathbb{E}[d_t] > 0$, and DM rarely favours it
- ▣ **Clark–West** (CW, Clark & West (2007)): one-sided HAC t -test of $\mathbb{E}[f_t] = 0$,

$$f_t = (y_t - \hat{p}_{0,t})^2 - [(y_t - \hat{p}_{1,t})^2 - (\hat{p}_{0,t} - \hat{p}_{1,t})^2]$$

- ▶ derived for nested linear models estimated by OLS; for penalised-logit probabilities an approximation, not a theorem
- ▣ **Giacomini–White** (GW, Giacomini & White (2006)): tests the forecasting *method* as deployed, with a bounded estimation window; CW asks whether the population model adds information (statistic in the Appendix)

Feature Importance: MDI, MDA and SHAP

Mean decrease impurity (MDI)

- Average Gini reduction from splits on a feature, over the trees
- In sample, trees only; favours continuous, high-cardinality features

Mean decrease accuracy (MDA)

- Drop in an out-of-sample score (here AUC) when a feature is randomly permuted
- Any model; can be negative: the feature hurts

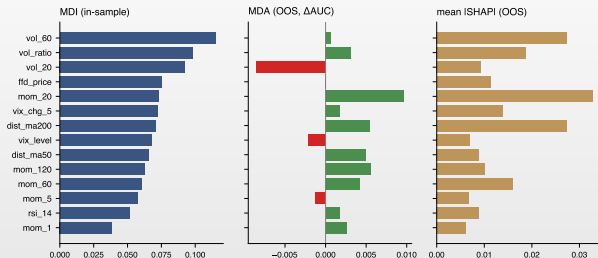
SHAP (Lundberg & Lee (2017))

- Shapley value ϕ_j : average marginal contribution of feature j over coalitions
- Additive: $\hat{f}(x) = \phi_0 + \sum_j \phi_j$; mean $|\phi_j|$ = global importance

- MDI answers “what did the model use?”; MDA and SHAP answer “what helps on new data?”
- Correlated features share and dilute importance (substitution): cluster them first; coalition formula and the conditional vs. interventional choice in the Appendix

 Shapley Values

Feature Importance in the Case Study



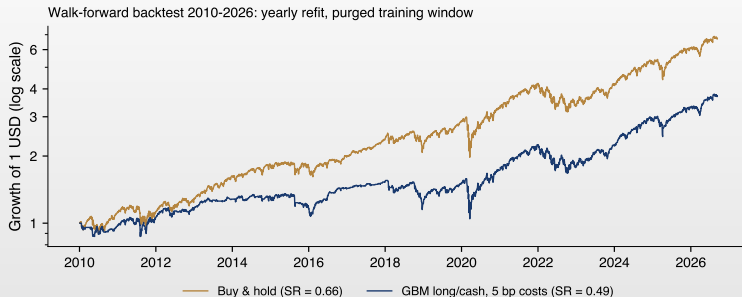
- Fit on the first 70%, evaluated on the last 30% after a purge; **RF** gives MDI and MDA, **GBM** gives SHAP: the differences are not purely methodological
- MDI ranks volatility first; MDA and SHAP point to `mom_20` and `dist_ma200`
- `vol_20`: high MDI, **negative** MDA: a sign of overfitting, not a proof
- All importances are tiny ($\Delta AUC < 0.01$)
- Correlated features: permutation creates off-manifold points, so MDA and SHAP extrapolate (Hooker, Mentch & Zhou (2021))

 **MFM_ch13_feature_importance**

Walk-Forward Backtest: Strategy and Design

- ▣ Position $w_t \in \{0, 1\}$: long the S&P 500 if $\hat{p}_t > 0.5$, otherwise cash, decided with information up to t
- ▣ Strategy return $r_{t+1}^s = w_t r_{t+1} - c |w_t - w_{t-1}|$, r_{t+1} the daily log return; $r - c$ is the first-order approximation of $\log(e^r - c)$
 - ▶ turnover $|w_t - w_{t-1}|$; cost $c = 5$ bp (0.05%) per unit traded; exposure $\bar{w} = \frac{1}{T} \sum_t w_t$
- ▣ Model: GBM refitted every January on all past data, purging labels that overlap the test year
- ▣ Out-of-sample period 2010–2026, one refit a year; benchmark buy-and-hold ($w_t = 1$: exposure 100%, no turnover)
- ▣ The only decision the backtest evaluates is the one a trader could have taken on day t

Walk-Forward Backtest: Results



- 2010–2026: strategy Sharpe **0.49** vs. buy-and-hold **0.66** (annualised); average exposure 77%; the model mostly learned “be long” and exited at the wrong times
- Paired difference -0.17 (Lo SE ≈ 0.25 each): Ledoit–Wolf HAC SE 0.095, $p = 0.073$ (Ledoit & Wolf (2008)); block-bootstrap CI $[-0.36; 0.02]$: no evidence of skill, and no significant evidence that the strategy is worse
- With $d = 0.15$, chosen by the same rule on data up to 2009, SR = 0.43; without ffd_price 0.51

Walk-Forward: Do the Forecasts Beat a Constant?

- ▣ Walk-forward 2010–2026, $n = 4,198$ days, 60.2% “up”; benchmark: the constant forecast = training share of “up”
- ▣ Logit (nests the constant): log-loss 0.6822 vs. 0.6757
 - ▶ DM $t = 2.39$ ($p = 0.017$): the estimated logit is significantly *worse*
 - ▶ Clark–West $t = -0.21$ (one-sided $p = 0.58$): no evidence that the features carry information
 - ▶ Giacomini–White = 5.77 (χ^2_2 , $p = 0.056$): no significant difference at 5%; approximate, since the window expands (rolling-window result in the Appendix)
- ▣ GBM: log-loss 0.6971; DM $t = 5.09$, GW = 25.8 ($p < 0.001$): worse than a constant; Pesaran–Timmermann -0.74
- ▣ An honest negative result is a result: no model beats “always up” or a constant forecast

 inference_tests.py

Selection Bias and the False Strategy Theorem

- ▣ A researcher tries N strategies or configurations and reports the best; even if all have zero true Sharpe ratio, $\max_n \widehat{SR}_n$ grows with N : **selection bias**
- ▣ Harvey, Liu & Zhu (2016): given hundreds of published factors, a new factor needs $t > 3$, not 2

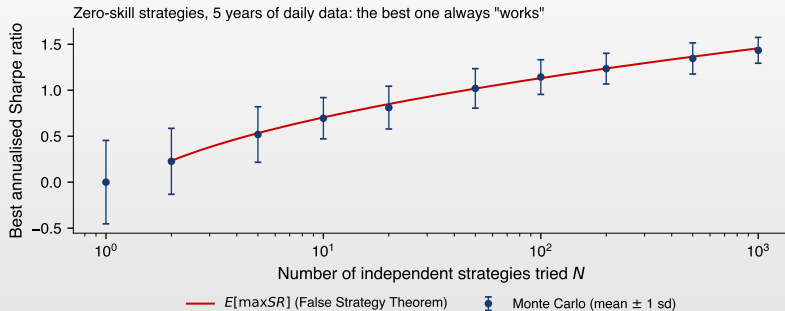
Theorem 1 (False Strategy Theorem, Bailey & López de Prado (2014))

- ▣ If $\widehat{SR}_n$, $n = 1, \dots, N$, are independent and approximately $N(0, \text{Var}[\widehat{SR}])$ (no skill), the **expected maximum Sharpe ratio** is

$$SR_0 \equiv \mathbb{E} \left[\max_n \widehat{SR}_n \right] \approx \sqrt{\text{Var}[\widehat{SR}]} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

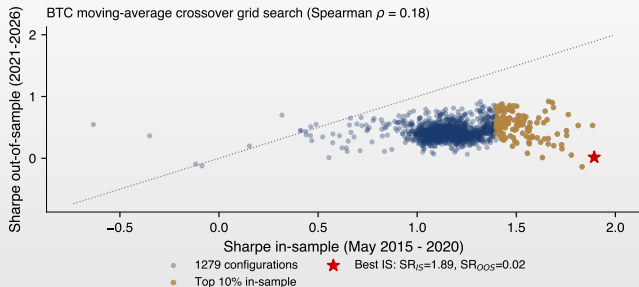
- ▣ $\gamma \approx 0.5772$: Euler–Mascheroni constant; Φ : standard Normal distribution function; i.i.d. returns: $\text{Var}[\widehat{SR}] \approx 1/T$ per period

The Best of N Zero-Skill Strategies



- Strategies with i.i.d. zero-mean returns, 5 years of daily data (annualised $\text{Var}[\widehat{SR}] \approx 1/5$)
- Trying $N = 1000$ variants yields a best Sharpe ratio of ≈ 1.45 — **with no skill at all**
- The Monte Carlo simulation agrees with the theorem

In-Sample vs. Out-of-Sample: Bitcoin Grid Search



- Moving-average (MA) crossover on Bitcoin: long when the fast MA of the log price exceeds the slow MA by more than a band; 1,279 configurations (fast, slow, band); in sample 15 May 2015 – 2020, out of sample 2021–2026
- Selected rule (14/20 days): SR_{IS} **1.89** → SR_{OOS} **0.02** (annualised with $\sqrt{365}$); Spearman rank correlation IS–OOS $\rho = 0.18$
- Median OOS SR 0.55 for the top 10% in sample vs. 0.42 for all: the *family* of trend rules carries information, the *selected* optimum is noise

Sharpe Ratio Inference: Standard Error and PSR

- Per-period Sharpe ratio (SR) $\widehat{SR} = \bar{r}/\hat{\sigma}$ of daily log returns, risk-free rate 0; skewness $\gamma_3 = \mathbb{E}[(r - \mu)^3]/\sigma^3$, ordinary (not excess) kurtosis $\gamma_4 = \mathbb{E}[(r - \mu)^4]/\sigma^4$
- i.i.d. returns, delta method (Lo (2002); Opdyke (2007)): $\text{Var}(\widehat{SR}) \approx (1 - \gamma_3 \widehat{SR} + \frac{\gamma_4 - 1}{4} \widehat{SR}^2) / T$; Normal distribution: $(1 + \frac{1}{2} \widehat{SR}^2) / T$

Definition 5 (Probabilistic Sharpe Ratio (PSR), Bailey & López de Prado (2012))

- Probability that the true SR exceeds a benchmark SR^* , the variance above in the denominator:

$$\widehat{PSR}(SR^*) = \Phi \left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- Annualising with $\sqrt{q}$ needs zero autocorrelation; otherwise $SR_q = \eta(q) SR$, $\eta(q) = q / \sqrt{q + 2 \sum_{k=1}^{q-1} (q-k) \rho_k}$ (Lo (2002))
- Two strategies on the same days: test $SR_a - SR_b$ with the HAC delta method or a studentised block bootstrap (Ledoit & Wolf (2008)), not with two separate intervals

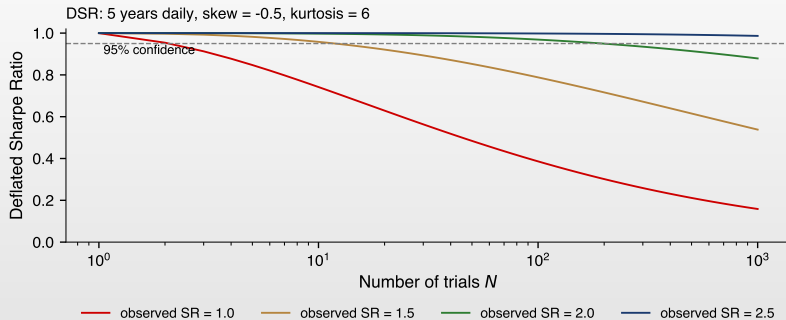
Deflated Sharpe Ratio: the Best Bitcoin Rule

- ▣ **Deflated Sharpe Ratio** (DSR, Bailey & López de Prado (2014)): $DSR = \widehat{PSR}(SR_0)$, with SR_0 the expected maximum of the False Strategy Theorem computed with $\text{Var}[\widehat{SR}_n]$ of the trials
 - ▶ corrects for selection (N trials), non-Normality and sample length; reject unless $DSR > 0.95$, and report N
- ▣ Worked example, best in-sample rule: $T = 2,058$ days, $\widehat{SR} = 0.0991$ a day (1.89 annualised), $\hat{\gamma}_3 = 0.12$, $\hat{\gamma}_4 = 12.33$
 - ▶ denominator $\sqrt{1 - 0.12 \cdot 0.0991 + \frac{12.33-1}{4} \cdot 0.0991^2} = 1.008$; $N = 1,279$ trials

$\text{Var}[\widehat{SR}_n]$ from	SD	SR_0 (per day)	z / DSR
dispersion of the 1,279 trial SRs	0.0106	0.035	2.88 / 0.998
independent trials, $1/T$	0.0220	0.073	1.16 / 0.877

- ▣ The trials are highly correlated, so their dispersion understates the search: the verdict depends on this choice, and the out-of-sample SR of 0.02 sides with the conservative value

Deflated Sharpe Ratio in Practice



- 5 years of daily returns, skewness -0.5 , kurtosis 6
- An annualised SR of 1.0 is significant only if (almost) no other strategy was tried: it fails after 2–3 trials
- $SR = 1.5$ survives ≈ 10 trials, $SR = 2.0 \approx 200$; only $SR = 2.5$ survives 1000

Data-Snooping Tests on the Bitcoin Grid

- K rules, differentials $d_{k,t} = r_{k,t} - r_{0,t}$ against a benchmark r_0 ; $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$; the null distribution comes from a **stationary bootstrap** (Politis & Romano (1994)) of whole days, which keeps the correlation between rules: no “effective” N needed
 - ▶ Reality Check (RC, White (2000)): $\max_k \sqrt{n} \bar{d}_k$; Superior Predictive Ability (SPA, Hansen (2005)): studentised, clearly bad rules not recentred; Romano–Wolf StepM (Romano & Wolf (2005)): names *which* rules win, with family-wise error rate (FWER) control
- 1,279 rules, in-sample $n = 2,058$ days, mean block 20 days, 2,000 replications (implementation in the Appendix)

Benchmark	largest studentised mean	RC p	SPA p	StepM rules kept
cash ($r_0 = 0$)	–	0.004	0.002	314
buy-and-hold	0.88 (the 14/20 rule)	0.74	0.70	0

- Against cash, some trend rule earns a positive mean: Bitcoin’s own trend; no rule beats holding Bitcoin, even in sample, consistent with the OOS SR of 0.02
- DSR asks “above the luck of N trials?” and needs N and $\text{Var}[\widehat{SR}]$ (0.998 or 0.877); SPA takes both from the data

 inference_tests.py

Probability of Backtest Overfitting (PBO)

Combinatorially symmetric cross-validation (CSCV): Bailey, Borwein, López de Prado & Zhu (2017)

1. Matrix of returns: T periods $\times$ N configurations; split the rows into S contiguous blocks
 2. For each of the $\binom{S}{S/2}$ choices, half of the blocks are in sample (IS), the rest out of sample (OOS)
 3. Pick the best IS configuration n^* ; rank all N configurations OOS in ascending order and set $\bar{\omega} = \text{rank}(n^*) / (N + 1) \in (0, 1)$
 4. Logit $\lambda = \log \frac{\bar{\omega}}{1 - \bar{\omega}}$; $PBO = \Pr[\lambda \leq 0]$: the share of splits in which the IS winner ranks at or below the OOS median
- ▣ $PBO \approx 0$: the IS winner reliably stays above the OOS median; $PBO \approx 0.5$ or more: selecting the IS winner is no better than picking at random
 - ▣ PBO evaluates the **research procedure**, not one strategy, and is not the probability of losing money; like DSR it needs **every** trial logged (Seminar B11)

Selection Is Not Inference: Post-Double Selection

- ▣ The LASSO path answers “which feature enters first”, not “which feature matters”; OLS t -statistics after LASSO ignore the selection step
- ▣ A control correlated with the regressor of interest d_t but only weakly with y_t gets dropped: its effect moves into the coefficient of d_t (omitted-variable bias)
- ▣ Partially linear model: $y_t = \theta d_t + g(x_t) + \varepsilon_t$, $d_t = m(x_t) + v_t$, with g , m approximately sparse
- ▣ Post-double selection (Belloni, Chernozhukov & Hansen (2014)):
 1. LASSO of y_t on $x_t \Rightarrow$ selected set $\hat{S}_y$
 2. LASSO of d_t on $x_t \Rightarrow$ selected set $\hat{S}_d$
 3. OLS of y_t on d_t and $x_{\hat{S}_y \cup \hat{S}_d}$, robust standard error
- ▣ A control is missed only if it matters for *neither* equation, so the omitted bias is second order; valid under independent sampling and sparsity, with HAC errors an approximation for time series (penalty and conditions in the Appendix)

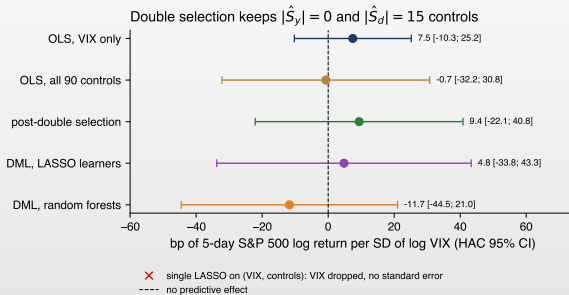
Double/Debiased ML: Orthogonal Scores and Cross-Fitting

- ▣ Chernozhukov et al. (2018): residualise both variables with *any* learner (LASSO, RF, boosting)

$$\tilde{y}_t = y_t - \hat{\ell}(x_t), \quad \tilde{v}_t = d_t - \hat{m}(x_t), \quad \hat{\theta} = \frac{\sum_t \tilde{v}_t \tilde{y}_t}{\sum_t \tilde{v}_t^2}$$

- ▣ Neyman-orthogonal score $\psi = (\tilde{y} - \theta \tilde{v}) \tilde{v}$: first-order errors in $\hat{\ell} = \hat{\mathbb{E}}[y | x]$ and $\hat{m}$ do not bias $\hat{\theta}$ (rate conditions in the Appendix)
- ▣ Cross-fitting: fit $\hat{\ell}$, $\hat{m}$ on $K - 1$ folds, residualise the held-out fold, repeat; removes the own-observation overfitting bias
- ▣ The published theory assumes independent sampling; with overlapping 5-day labels we use contiguous folds purged by 5 days around the test fold and HAC standard errors, a data limitation stated as such, not a new estimator

Worked Example: Does the VIX Predict S&P 500 Returns?



- y_t : 5-day forward log return (bp); d_t : log VIX, standardised (1 SD = 0.356); 90 controls: 12 technical features, squares, pairwise products; $n = 6,269$; HAC errors, 4 lags
- The single LASSO drops the VIX and every control: “no effect”, with no standard error
- Double selection keeps $|\hat{S}_y| = 0$, $|\hat{S}_d| = 15$; 95% CI $[-22.1; 40.8]$ bp per week per SD
- Honest intervals contain zero and large effects of both signs: the data cannot decide

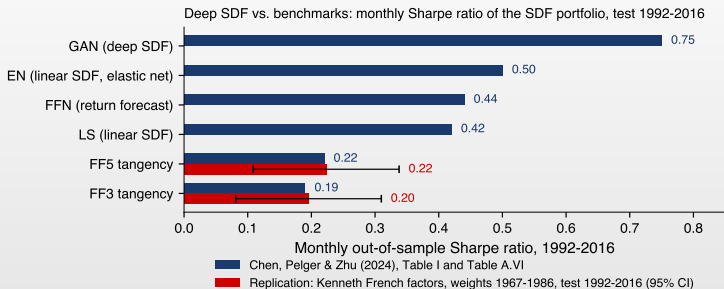
The Virtue of Complexity, and a Critical Reading

- ▣ Kelly, Malamud & Zhou (2024): ridge regressions on P random Fourier features of a few predictors, rolling windows as short as $T = 12$ months, P in the thousands
 - ▶ theory: with $P > T$ the minimum-norm solution shrinks implicitly ("double descent"); R^2_{OOS} may be negative while the Sharpe ratio of market timing **risers** with complexity; complexity is not automatically overfitting
- ▣ Nagel (2025): for $P \gg T$ the forecast is a similarity-weighted average of the T training returns; in short windows similarity is mostly recency, so the method becomes a **volatility-timed momentum** strategy, which loses on artificial data with reversals
- ▣ Avramov, Cheng & Metzker (2023): deep-learning signal profits come from hard-to-arbitrage stocks; excluding microcaps, distressed firms or high-volatility episodes attenuates them, and costs erode them further
- ▣ Chen, Pelger & Zhu (2024): impose economic structure instead; the no-arbitrage moment conditions become the loss of a deep network (next slides)
- ▣ Survey of the field: Kelly & Xiu (2023)

Case Study: Deep Learning in Asset Pricing, Question and Method

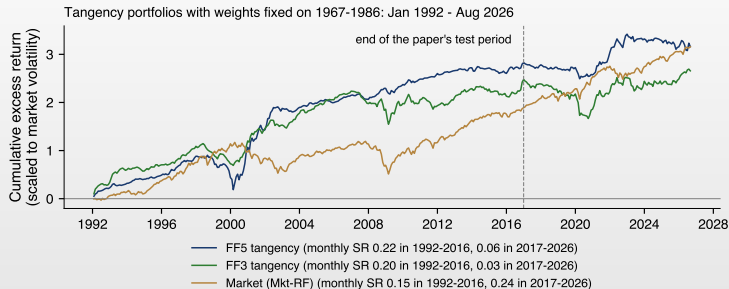
- [Chen, Pelger & Zhu \(2024\)](#): estimate the stochastic discount factor (SDF) M_{t+1} that prices **all individual US stocks**, conditioning on 46 firm characteristics and on the history of 178 macroeconomic series
 - ▶ no arbitrage: $E_t[M_{t+1}R_{t+1,i}^e] = 0$ for every excess return; with $M_{t+1} = 1 - \omega_t^\top R_{t+1}^e$, $F_{t+1} = \omega_t^\top R_{t+1}^e$ is the conditional tangency portfolio and $E_t[R_{t+1,i}^e] = \beta_{t,i} E_t[F_{t+1}]$ (Section I.A, eq. 1)
- Why: forecasting ([Gu, Kelly & Xiu \(2020\)](#)), the benchmark feedforward network, FFN) fits a second moment where signal-to-noise is low; the no-arbitrage condition targets a first moment, and averaging over time removes much of the noise
- Method: adversarial generalized method of moments (GMM), $\min_{\omega} \max_g \frac{1}{N} \sum_j (E[M_{t+1}R_{t+1,j}^e g(l_t, l_{t,j})])^2$ (eq. 3, after [Hansen & Jagannathan \(1997\)](#)); an FFN for ω , a long short-term memory network (LSTM) that turns the macro history into 4 states, an adversary g that builds the hardest test assets; the full model is labelled GAN
- Chronological split: training 1967–1986, validation 1987–1991, test 1992–2016; configurations ranked by the validation Sharpe ratio (data and tuning in the Appendix)

Deep Learning in Asset Pricing: Sharpe Ratios, Paper and Replication



- Blue: the paper, monthly out-of-sample SR 1992–2016: GAN 0.75, elastic net linear SDF (EN) 0.50, FFN 0.44, unregularised linear SDF (LS) 0.42 (Table I); FF5 0.22, FF3 0.19 (Table A.VI); the regularised no-arbitrage model already beats the forecasting network
- Red: FF3 and FF5 tangency portfolios $w \propto \hat{\Sigma}^{-1} \hat{\mu}$ on the [Kenneth French](#) factors, weights fixed on 1967–1986: FF5 0.22 [0.11; 0.34], FF3 0.20 [0.08; 0.31] (SE 0.058, [Lo \(2002\)](#)): both published values lie inside the intervals

Deep Learning in Asset Pricing: the Benchmarks after 2016



- Same fixed weights, January 1992 – August 2026, each series scaled to the volatility of the market excess return on 1992–2016; monthly SR 2017–2026 ($T = 116$, $SE \approx 0.09$): FF5 0.06, FF3 0.03, market 0.24; on 1992–2016: 0.22, 0.20, 0.15
- Replicable with public data: the Fama–French benchmarks; not replicable: the stock-level GAN, LS and EN (the CRSP panel is not in the course data); the post-2016 fall is a warning, not yet a test, and the same question applies to a GAN frozen in 1991 (details in the Appendix)

An Open Question: Discovery or Data Mining at Scale?

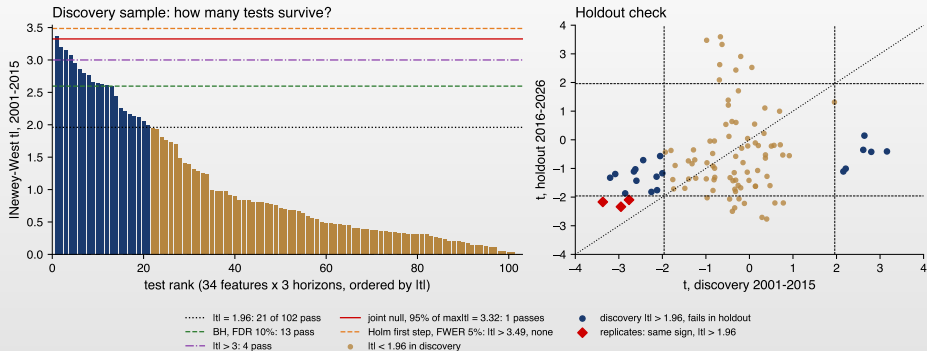
- ▣ [Chen, Lopez-Lira & Zimmermann \(2022\)](#): mining 29,000 accounting ratios for $t > 2$ gives cross-sectional predictability similar to peer-reviewed predictors; for both, about half remains after the original samples
- ▣ [Novy-Marx & Velikov \(2025\)](#): an LLM (large language model) pipeline mines over 30,000 signals, keeps 96 that pass a fixed protocol and writes hundreds of complete papers, with invented motivations and some imagined citations: HARKing (hypothesising after the results are known) at scale
- ▣ Open: when a machine proposes features, model and story, what is the number of trials, and what separates a discovery from the best of many draws?
- ▣ The discovery loop: literature $\rightarrow$ competing hypotheses $\rightarrow$ data and code $\rightarrow$ robustness fixed in advance $\rightarrow$ referee-style critique $\rightarrow$ report every test (reusable prompts in the Appendix)
- ▣ What the human checks: every reference resolves (DOI, arXiv); every trial is logged, so M in the correction counts them all; correlated tests get a joint null; transformations live inside the training fold

Mini-Case: 102 Tests, One Old Survivor

- A typical assistant proposal: “add more indicators: momentum, moving-average distances, volatility ratios, RSI and VIX changes over several windows”
 - ▶ 34 features $\times$ 3 horizons (1, 5, 20 days) = 102 predictive regressions on 2001–2015, Newey–West t ; $H_{0,j} : \beta_j = 0$
- $|t| > 1.96$: **21** of 102, against 5.1 expected under the global null at 5%
- Holm (FWER 5%): **0**; Benjamini–Hochberg (BH, false discovery rate (FDR) 10%, [Benjamini & Hochberg \(1995\)](#)): **13**; $|t| > 3$ ([Harvey, Liu & Zhu \(2016\)](#)): **4**
- Joint null (500 circular shifts of the target): 95% quantile of $\max_j |t_j| = 3.32$; one survivor, the 1-day return on the next day's return, $t = -3.37$: short-term reversal, a known effect
- Holdout 2016–2026: 3 of the 21 keep their sign with $|t| > 1.96$, all reversal variants; the VIX-based “finds” fail
- Holm and BH control different error rates at different levels; BH needs independence or positive dependence of the true-null p-values, which correlated predictors do not ensure

 [ai_discovery_case.py](#)

Mini-Case: Thresholds and the Holdout



- Left: the 102 discovery $|t|$ in decreasing order; each threshold answers a different question, and only the joint null accounts for the correlation between the tests
- Right: discovery vs. holdout t ; the three replications (red) are all short-term reversal; the significant VIX features collapse towards zero

Your Turn: a Project Seed

- ▣ **Question:** can an AI-assisted feature search find a predictor of index returns (S&P 500, BET, Bitcoin) that survives a pre-registered correction and a holdout, beyond short-term reversal?
- ▣ **Replicate first:** this mini-case (21 of 102, one survivor at the joint null)
- ▣ **Deliverable:** a feature list fixed before the data are touched; all tests with Holm, BH and the max- $|t|$ null; a holdout; DSR and PBO for any strategy built on the survivors
- ▣ **Log:** each prompt that shaped a decision in AI_USE.md; each invented reference, leak or undercounted trial in AI_ERRORS.md, with the test that caught it ([templates](#))

Key Takeaways and a Decision Checklist

Three empirical lessons

- An ADF rejection is not stationarity: truncated FFD of the S&P 500 log price stays $I(1)$; estimate memory with local Whittle
- Shuffled K -fold reports 69% on pure noise; on the S&P 500 no model beats “always up” (PT, DM, GW)
- The best of many backtests must be deflated or bootstrapped: Bitcoin grid 1.89 in sample, 0.02 out of sample

Next

- Chapter 14: deep learning and time-series foundation models; Chapter 15: large language models and sentiment

Before any claim

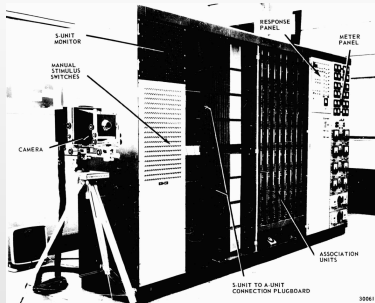
- Point-in-time data; transformations fitted on training folds only
- Volatility-scaled labels with recorded spans
- Purged K -fold or CPCV with embargo; walk-forward for the final test
- A naive baseline; PT, DM / CW / GW and block-bootstrap intervals
- Costs included; every trial logged; DSR, SPA and PBO
- “Selected” or “important” is not “significant”: double selection or DML for inference

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

-  Random Forests
-  Shapley Values
-  Shapley Curves: A Smoothing Perspective
-  Machine Learning in Financial Risk
-  Introduction to Reinforcement Learning
-  Generative Adversarial Networks
-  MSCA Digital: The Need for Explainable AI
-  Business Analytics and Data Science
-  Model Selection Criteria
-  PAC Learning

Appendix: From the Perceptron to Deep Blue



Mark I Perceptron, operator's manual (1960)

Figure: J.C. Hay, A.E. Murray (1960); public domain; Wikimedia Commons



IBM Deep Blue, Computer History Museum

Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons

- **Perceptron:** a learning rule for a single artificial neuron [Rosenblatt \(1958\)](#); the Mark I was hardware built to recognise simple images
- 1997: IBM's Deep Blue beats world chess champion Garry Kasparov [Campbell, Hoane & Hsu \(2002\)](#)
- Deep Blue searched with hand-built rules; modern ML (this chapter) *learns* the rules from data

Appendix: the In-Sample R^2 of a Predictor without Skill

- A monthly R^2 of 0.5% is economically large, but sampling noise is of the same order
- In-sample OLS R^2 of one predictor with no skill (intercept, independent errors, $n \approx 300$ months): $nR^2 \approx \chi_1^2$
 - ▶ so R^2 has mean $1/n \approx 0.3\%$ and standard deviation $\sqrt{2}/n \approx 0.5\%$
- R_{OOS}^2 can be negative, has no such null distribution and needs its own test (Clark–West)

Appendix: Worked Example: How a Forest Averages Its Trees

- Random forest of the case study (400 trees, minimum leaf size 100), trained on the first four purged folds
- Prediction for the last observation (11 September 2026): $P(\text{up in 5 days})$ from each tree
 - ▶ individual trees range from 0.31 to 0.88 (standard deviation 0.11)
 - ▶ 81.5% of the trees vote “up” ($p > 0.5$)
 - ▶ forest probability = average of the 400 trees = 0.608
- Single trees disagree strongly; the average is far more stable
- Stability is not accuracy: the forest still does not beat “always up” (case study)

Appendix: Sampling Financial Data: Bars

Time and tick bars

- ▣ A **bar** = one row (open, high, low, close, volume) summarising a sampling interval
- ▣ Time bars: every fixed interval (1 minute, 1 day); oversample quiet periods; fat-tailed returns
- ▣ Tick bars: one bar every N transactions

Volume, dollar and information bars

- ▣ One bar every V shares or every D dollars traded: sampling by **information arrival**
- ▣ Returns closer to i.i.d. draws from the Normal distribution: [Clark \(1973\)](#); [Ané & Geman \(2000\)](#)
- ▣ Imbalance / runs bars: sample when order flow becomes one-sided

- ▣ This chapter uses daily time bars; tick data (crypto exchanges) in Chapters 10 and 16

Appendix: Fractional Weights and Their Asymptotics

Definition 6 (Fractional difference operator)

- With the backshift operator $BX_t = X_{t-1}$ and a real order d :

$$(1 - B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k, \quad \tilde{X}_t = \sum_{k=0}^{\infty} w_k X_{t-k}$$

- Weights: $w_0 = 1$, $w_k = -w_{k-1} \frac{d-k+1}{k} = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)} \sim \frac{k^{-(1+d)}}{\Gamma(-d)}$ as $k \rightarrow \infty$
- $d = 1$: $w = (1, -1, 0, 0, \dots)$ — ordinary returns
- $0 < d < 1$: weights decay hyperbolically $\Rightarrow$ long memory is kept
- Infinite sum: $\sum_{k=0}^{\infty} w_k = (1 - 1)^d = 0$, so the level of X_t cancels only in the limit

Appendix: Local Whittle and Exact Local Whittle

- Periodogram $I(\lambda_j)$ at $\lambda_j = 2\pi j/n$; local Whittle ([Robinson \(1995\)](#)) on the first m frequencies:

$$\hat{\delta} = \arg \min_{\delta} \log \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2\delta} I(\lambda_j) \right) - \frac{2\delta}{m} \sum_{j=1}^m \log \lambda_j, \quad \sqrt{m}(\hat{\delta} - \delta) \xrightarrow{d} N(0, \frac{1}{4})$$

- Exact local Whittle (ELW, [Shimotsu & Phillips \(2005\)](#); unknown mean: [Shimotsu \(2010\)](#)): periodogram of $(1 - B)^{\delta} X_t$, valid also for non-stationary $\delta \geq \frac{1}{2}$
- Why the FFD series looks like $\hat{\delta} \approx 0.75$: above frequency $1/K$ the truncated filter acts as $(1 - B)^{0.25}$; only $n/(2\pi K) \approx 2$ of the $m = 294$ frequencies lie below $1/K$, where the unit root returns
 - the estimate is the apparent memory over the band, not the integration order of the series

Appendix: Worked Example: Purging with 10 Observations

- ▣ Observations $t = 0, \dots, 9$; label of t ends at $t + 2$ (capped at 9)
- ▣ 5 contiguous folds of 2; embargo = 1 observation

Test	Removed	Train
0, 1	2, 3	4–9
2, 3	0, 1, 4, 5	6–9
4, 5	2, 3, 6, 7	0, 1, 8, 9
6, 7	4, 5, 8, 9	0–3
8, 9	6, 7	0–5

- ▣ Test $\{4, 5\}$: labels of 2 and 3 end at 4 and 5 $\Rightarrow$ purged; 6 is the embargo; 7 starts when the last test label ends $\Rightarrow$ purged
- ▣ With overlapping labels, up to 4 of 10 observations leave the training set

Appendix: When Is K-Fold Valid for Time Series?

- Bergmeir, Hyndman & Koo (2018): for a **purely autoregressive** model $y_t = g(y_{t-1}, \dots, y_{t-p}) + \varepsilon_t$ with serially uncorrelated errors, standard K -fold CV is valid
 - ▶ the errors in the test fold are uncorrelated with the training errors, so the CV error is unbiased for the OOS error
- It fails when the **errors** are serially correlated
 - ▶ overlapping labels: h -day forward returns share $h - 1$ daily returns
 - ▶ persistent features that are not in the model's lag set (rolling means, levels such as `ffd_price`)
 - ▶ under-specified dynamics (too few lags)
- The leakage experiment isolates exactly this: 20-day labels and 20-day rolling features
- Check before choosing: the autocorrelation of the CV residuals; if it is significant, purge and embargo

Appendix: Walk-Forward and Combinatorial Purged CV

Walk-forward (WF)

- Train on $[0, T_k]$, test on $(T_k, T_{k+1}]$, roll forward
- Mimics real-time deployment; no future data in training
- But a **single path**: high variance, easy to overfit by repeated trials

Combinatorial purged CV (CPCV)

- N groups; test on all $\binom{N}{k}$ combinations of k groups, with purging and embargo
- $\varphi = \frac{k}{N} \binom{N}{k}$ backtest paths; e.g. $N = 6$, $k = 2$: 15 splits, 5 paths
- A **distribution** of Sharpe ratios instead of one number

- Rule of thumb: purged K-fold / CPCV for model selection; WF for the final, realistic evaluation

Appendix: Testing Directional Accuracy: Pesaran–Timmermann

- H_0 : the predicted sign x_t is independent of the realised sign y_t (Pesaran & Timmermann (1992))
- $\hat{P}$ = share of correct signs; under H_0 the expected share is $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$

$$PT = \frac{\hat{P} - \hat{P}^*}{\sqrt{\hat{V}(\hat{P}) - \hat{V}(\hat{P}^*)}} \xrightarrow{d} N(0, 1), \quad \hat{V}(\hat{P}) = \frac{\hat{P}^*(1 - \hat{P}^*)}{n}$$

$$\blacktriangleright \hat{V}(\hat{P}^*) = [(2\hat{p}_y - 1)^2 \hat{p}_x(1 - \hat{p}_x) + (2\hat{p}_x - 1)^2 \hat{p}_y(1 - \hat{p}_y)]/n + 4\hat{p}_y \hat{p}_x(1 - \hat{p}_y)(1 - \hat{p}_x)/n^2$$

- Logit above: $\hat{p}_y = 0.581$, $\hat{p}_x = 0.983 \Rightarrow \hat{P}^* = 0.578 > \hat{P} = 0.576$; $PT = -1.28$, one-sided $p = 0.90$
- PT conditions on the marginal frequencies automatically: no need to guess the baseline
- Overlapping 5-day labels: PT assumes independent days; the same null as the slope t -test in $y_t = a + b x_t + u_t$ with Newey–West errors (Newey & West (1987)): $t = -1.00$

Appendix: ROC AUC and Its Sampling Uncertainty

- Receiver operating characteristic (ROC) curve: true positive rate $TP/(TP + FN)$ against false positive rate $FP/(FP + TN)$ over all thresholds
- Area under the curve (AUC) = $P(\hat{p}_{\text{pos}} > \hat{p}_{\text{neg}}) + \frac{1}{2}P(\hat{p}_{\text{pos}} = \hat{p}_{\text{neg}}) = U/(n_1 n_0)$, the Mann–Whitney statistic; 0.5 = no ranking ability
- DeLong standard error (DeLong, DeLong & Clarke-Pearson (1988)): from the placement values of each observation; assumes independent observations
- Circular block bootstrap (Künsch (1989)): resample blocks of $L = 20$ consecutive days, 1,000 replications; keeps serial dependence
- Logit, pooled out-of-fold AUC = 0.479
 - ▶ DeLong 95% CI [0.465; 0.494]: “significantly worse than random”
 - ▶ block bootstrap 95% CI [0.455; 0.504]: noise around 0.5
- An AUC below 0.5 is anti-ranking only if it survives a test that respects the dependence

Appendix: Giacomini–White, Statistic and Rolling Window

- **Giacomini & White (2006)**: the null concerns the forecasting **method** (model + estimation window + refitting rule), not the population model

$$H_0 : \mathbb{E}[d_{t+\tau} \mid \mathcal{F}_t] = 0 \Rightarrow \mathbb{E}[h_t d_{t+\tau}] = 0, \quad GW = n \bar{\mathbf{Z}}^\top \hat{\Omega}^{-1} \bar{\mathbf{Z}} \xrightarrow{d} \chi_q^2$$

- ▶ $\mathbf{Z}_t = h_t d_{t+\tau}$; test functions h_t (here 1 and the lagged loss differential, $q = 2$); $\tau = 5$ -day horizon
- The theory requires a bounded (rolling) estimation window, so that estimation noise never vanishes; an expanding window lies outside the theorem
- CW: does the *population* model with features add information? GW: does *my estimated* rule, as deployed, forecast better?
- S&P 500 logit, rolling 8-year window, as the GW theory requires: $GW = 26.5$ ($p < 0.001$): the logit forecasts worse than the constant

 inference_tests.py

Appendix: Feature Importance: SHAP

Definition 7 (Shapley value of feature j (SHAP, Lundberg & Lee (2017)))

- Average marginal contribution of j over all coalitions $S \subseteq \{1, \dots, p\} \setminus \{j\}$:

$$\phi_j = \sum_S \frac{|S|! (p - |S| - 1)!}{p!} [v(S \cup \{j\}) - v(S)], \quad \hat{f}(x) = \phi_0 + \sum_j \phi_j$$

- Interventional SHAP: $v(S) = \mathbb{E}[\hat{f}(x_S, X_{\bar{S}})]$, averaging $\hat{f}$ with the features outside S drawn from a background sample (as in the Quantlet); $\phi_0 = v(\emptyset) =$ average prediction

- Conditional SHAP uses $v(S) = \mathbb{E}[\hat{f}(X) \mid X_S = x_S]$; the two differ when features are dependent
- Additive attribution of each prediction; mean $|\phi_j|$ over observations = global importance
- **Substitution effects**: correlated features share (and dilute) importance — cluster them first



Appendix: Reading Feature Importances

- ▣ **MDI** ranks volatility features first; **MDA** and **SHAP** point to `mom_20` and `dist_ma200`
- ▣ `vol_20`: high MDI but **negative** MDA — a sign of overfitting, not a proof: sampling noise and correlated features also move MDA
- ▣ Importances are tiny ($\Delta\text{AUC} < 0.01$), consistent with the model comparison
- ▣ Permuting one of several correlated features creates points off the data manifold: MDA and SHAP then **extrapolate** (Hooker, Mentch & Zhou (2021)); permute within clusters or refit without the feature
- ▣ Importances are estimates and need their own uncertainty analysis; honest trees give asymptotically Normal forest *predictions* and CIs (Wager & Athey (2018)), not CIs for importances
- ▣ **Backtesting is not a research tool**, importance analysis is: López de Prado (2018)

Appendix: Data-Snooping Tests: Reality Check, SPA, Romano–Wolf

- K rules, performance differentials $d_{k,t} = r_{k,t} - r_{0,t}$ against a benchmark r_0 ; $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$
- Reality Check (RC, [White \(2000\)](#)): statistic $\max_k \sqrt{n} \bar{d}_k$; null distribution from the **stationary bootstrap** ([Politis & Romano \(1994\)](#)) of the whole $n \times K$ matrix
 - ▶ resampling whole days keeps the correlation between rules: no need to count “effective” trials
- Superior Predictive Ability (SPA, [Hansen \(2005\)](#)): studentised $\max_k \sqrt{n} \bar{d}_k / \hat{\omega}_k$; clearly bad rules are not recentred, so adding junk rules does not destroy power
- Romano–Wolf StepM ([Romano & Wolf \(2005\)](#)): step-down procedure that names *which* rules beat the benchmark, with family-wise error rate (FWER) control
- DSR asks “is the best SR above the luck of N trials?” and needs N and $\text{Var}[\widehat{SR}]$; these tests take both from the data

Appendix: Double Selection and DML, Penalty and Conditions

- Partially linear model: $\mathbb{E}[\varepsilon_t \mid d_t, x_t] = 0$, $\mathbb{E}[v_t \mid x_t] = 0$, $\text{Var}(v_t) > 0$; g , m approximately sparse
- Post-double selection: the heteroskedasticity-robust standard error is valid under independent sampling, regularity conditions and sparsity $s^2 \log^2(p \vee n)/n \rightarrow 0$ (Belloni, Chernozhukov & Hansen (2014)); with time series, HAC errors are an approximation
- LASSO objective $\frac{1}{n} \sum_t (y_t - x_t^\top \beta)^2 + \frac{\lambda}{n} \sum_j \hat{\psi}_j |\beta_j|$, plug-in $\lambda = 2c\sqrt{n} \Phi^{-1}(1 - \gamma/(2p))$, $c = 1.1$, $\gamma = 0.1/\log n$, loadings $\hat{\psi}_j^2 = \frac{1}{n} \sum_t x_{tj}^2 \hat{\varepsilon}_t^2$, iterated
- DML (Chernozhukov et al. (2018)): with $\ell(x) = \mathbb{E}[y \mid x]$, it suffices (with consistency and regularity) that $\|\hat{m} - m\| (\|\hat{\ell} - \ell\| + \|\hat{m} - m\|) = o_P(n^{-1/2})$, e.g. both $o_P(n^{-1/4})$

Appendix: Machine Learning for Inference in Asset Pricing

- [Feng, Giglio & Xiu \(2020\)](#): does a new factor add pricing information given hundreds of existing ones?
 - ▶ double selection of the controlling factors from the zoo, then a valid t -test of the new factor's risk price
 - ▶ the inferential answer to [Harvey, Liu & Zhu \(2016\)](#), who raise the hurdle to $t > 3$
- Heterogeneous effects: honest causal forests give CIs for effects that vary with covariates ([Wager & Athey \(2018\)](#))
- Three rules
 - ▶ prediction and inference are different targets: a good predictor can give a biased $\hat{\theta}$
 - ▶ orthogonalise before you test, and cross-fit with folds that respect time
 - ▶ report CIs; “selected” or “important” is not “significant”

Appendix: Deep Learning for Sequences

Recurrent networks (LSTM, GRU)

- Hidden state carries memory: $h_t = g(h_{t-1}, x_t)$
- Gates control what is kept or forgotten, mitigating vanishing gradients
- Used for volatility, limit-order-book and intraday forecasting

Transformers

- Self-attention: $\text{softmax}(QK^\top / \sqrt{d_k})V$, with query, key, value matrices Q, K, V
- Long-range dependence, parallel training
- Backbone of large language models (Chapter 15)

Appendix: Time-Series Foundation Models

Pre-trained forecasters

- Pre-trained on large collections of series, used **zero-shot** (without re-training): Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), Moirai (Woo et al., 2024), Lag-Llama (Rasul et al., 2023)
 - Useful baselines for volatility and volume forecasting
 - For returns they inherit the same signal-to-noise limits
 - Pre-training data may overlap your test period: a new form of leakage
-
- Covered in depth, with the risk-forecast evaluation tools of Chapter 8, in Chapter 14

Appendix: Reinforcement Learning, Generative Models, Explainable AI, Risk

Reinforcement learning (RL)

- ▣ A policy $\pi(a | s)$ chooses actions a in states s to maximise cumulative reward
- ▣ Optimal execution, market making, deep hedging: Buehler et al. (2019)

Generative models

- ▣ GANs, variational autoencoders (VAEs), diffusion models: synthetic scenarios for stress tests

Explainable AI

- ▣ Required by regulators (EU AI Act, credit scoring, model risk)
- ▣ SHAP, LIME (local interpretable model-agnostic explanations), partial dependence

ML for risk

- ▣ Value-at-Risk (VaR): conditional quantiles (quantile forests, quantile-loss boosting); Expected Shortfall (ES): average of fitted tail quantiles or a joint VaR–ES loss; volatility: a squared-return target (Chapters 7–8)

▣ Quantinar:  Reinforcement Learning  GANs  Explainable AI  ML in Financial Risk

Appendix: Deep Learning in Asset Pricing, Data (Section III.A)

- Monthly returns of all CRSP stocks, January 1967 – December 2016; excess over the one-month Treasury-bill rate
- 46 firm characteristics in six groups: past returns, investment, profitability, intangibles, value, trading frictions (Table A.II)
 - ▶ Only stock-months with all 46 available: about 10,000 of 31,000 stocks; ranked each month into cross-sectional quantiles
- 178 macroeconomic series: 124 of FRED-MD ([McCracken & Ng \(2016\)](#)), the 46 monthly medians of the characteristics, 8 predictors of [Welch & Goyal \(2008\)](#); transformed to stationarity
- Split: training 1967–1986, validation 1987–1991, test 1992–2016

Appendix: Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C)

- Adversarial generalized method of moments: $\min_{\omega} \max_g \frac{1}{N} \sum_j (E[M_{t+1} R_{t+1,j}^e g(l_t, l_{t,j})])^2$ (eq. 3); g builds the test assets with the largest pricing errors, motivated by [Hansen & Jagannathan \(1997\)](#)
- Networks: an FFN for ω ; an LSTM that turns the macro history into 4 hidden states; the adversary g with 8 instruments; three steps: unconditional SDF, maximise over g , minimise again; the full model is labelled GAN (Section II.D)
- Tuning: 384 configurations ranked by the validation Sharpe ratio (SR); chosen: 2 layers of 64 units, dropout, Adam; ensemble of 9 fits (Table A.VIII); loadings: a second FFN fits $E_t[R_{t+1,i}^e F_{t+1}] \propto \beta_{t,i}$
- Benchmarks: the FFN of [Gu, Kelly & Xiu \(2020\)](#); the linear SDF $\omega_{t,i} = \theta^\top l_{t,i}$ on long and short characteristic legs, unregularised (LS) or elastic net (EN); FF3/FF5 tangency portfolios ([Fama & French \(2015\)](#))
- Metrics: SR of F ; $EV = 1 - \overline{\hat{e}^2} / \overline{(R^e)^2}$ (time averages of monthly cross-sectional means); cross-sectional R^2 : one minus mean squared time-averaged residual over mean squared time-averaged excess return, stocks weighted by T_i/T (II.F)

Appendix: Deep Learning in Asset Pricing, Results (1/2)

- Source: Chen, Pelger & Zhu (2024), *Management Science*
- Monthly out-of-sample SR of the SDF (Table I): GAN 0.75, EN 0.50, FFN 0.44, LS 0.42; FF5 0.22, FF3 0.19 (Table A.VI)
 - ▶ Reading: annualised 2.6 vs. 1.7 (EN) and 1.5 (FFN); the regularised linear no-arbitrage model (EN) already beats the forecasting network; unregularised LS does not
- Individual stocks (Table I): EV 8% (GAN) vs. 3–4%; cross-sectional R^2 23% vs. 14–19%
 - ▶ Reading: the no-arbitrage objective improves both fits, although it maximises neither
- Deciles sorted on the loadings β (Table II, test period, annualised): top minus bottom 0.39; FF5 alpha 0.39, $t = 9.96$; the GRS test rejects the CAPM, FF3 and FF5 ($p = 0.00$)
 - ▶ Reading: loadings predict returns, linearly ($R^2 = 0.97$ of mean return on β , Figure 8), and the factor models do not price them

Appendix: Deep Learning in Asset Pricing, Results (2/2)

- 46 sets of characteristic-sorted deciles (Table III): GAN cross-sectional R^2 above 90% for all; EV short-term reversal 0.70 vs. 0.43 (EN), momentum 0.54 vs. 0.26
 - ▶ Reading: gains come from extreme deciles and interactions; size and book-to-market deciles are easy for every model (Section III.E)
- Macro states (Figure 6): the 178 raw series make the out-of-sample SR collapse; GAN loses about 10% without macro states and 20% with unconditional moments only
 - ▶ Reading: the dynamics of macro data must be modelled (LSTM), not fed as the last increment
- Largest stocks (Table IV, monthly SR, models fitted on all stocks, evaluated on the subset): about 1,500 stocks GAN 0.41, FFN 0.30, EN 0.15; about 550 stocks 0.26, 0.24, 0.23; re-estimated on the 1,500: 0.42, 0.28, 0.42
 - ▶ Reading: the edge narrows sharply among the largest stocks; EN catches up when re-estimated on them

Appendix: Deep Learning in Asset Pricing, Replication and Extension

- Data limitation: the stock-level panel (CRSP returns, 46 characteristics) is not in the course data, so the GAN on individual stocks cannot be reproduced
- With the [Kenneth French Data Library](#) and FRED-MD, on the published split
 - ▶ Replicable: FF3 and FF5 tangency weights estimated on 1967–1986 (footnote 23), monthly SR on 1992–2016 next to Table A.VI
 - ▶ Not replicable with the course data: the published LS and EN (Tables I and III) use rank-weighted legs of 46 characteristics and stock-level betas
 - ▶ Analogue on portfolio data, an extension rather than a replication: the linear SDF ($\hat{\theta}_{LS}$ in closed form, $\hat{\theta}_{EN}$ with the objective of Section II.F) on halves (deciles 1–5, 6–10) of the 15 characteristics that the library sorts into deciles; its levels are not comparable with Tables I and III

Appendix: Deep Learning in Asset Pricing, Limits and an Open Extension

- Limitations stated by the authors
 - ▶ The positivity of the SDF is not imposed (footnote 9); only stock-months with all characteristics, which mainly drops very small stocks (Section III.A)
 - ▶ Frictions (Section III.I, Figure 17): annual SR 1.73 without the smallest 40% of stocks, 2.07 without the widest 40% bid–ask spreads, 1.87 without the lowest 40% turnover; see also [Avramov, Cheng & Metzker \(2023\)](#)
 - ▶ In-sample metrics exceed the out-of-sample ones, a sign of some overfitting (Section III.C); the hidden states are not comparable across fits, so they cannot be averaged (Section II.E)
- Open extension: freeze the 1967–1991 design and evaluate 2017–2026, after the paper's sample; does the linear no-arbitrage SDF still beat FF5, and do LSTM states of FRED-MD add to it?

Appendix: Deep Learning in Asset Pricing, Reading the Two Charts

- GAN 0.75 vs. FF5 0.22: with Lo SEs 0.065 and 0.058, the SE of the difference is at most 0.12 whatever their correlation, so the gap exceeds 4 SEs (i.i.d. Normal returns); a paired test on aligned returns ([Ledoit & Wolf \(2008\)](#)) is sharper
- The fixed FF5 weights load mainly on profitability and investment: RMW 0.42, CMA 0.30 (weights scaled to $\sum |w| = 1$); FF3 puts 0.61 on HML; both combinations were chosen on 1967–1986
- After 2016 the fixed tangency portfolios have monthly Sharpe ratios far below the market's; with $T = 116$ months the SE is about 0.09: a warning, not yet a test

 MFM_ch13_sdf_replication

Appendix: The Discovery Loop, Step by Step (1/2)

- **Literature:** "Summarise the evidence on how much published return predictability survives out of sample and after publication: DOI or arXiv id, sample, headline decay estimate."
 - ▶ Resolve every identifier; Semantic Scholar, Google Scholar, Elicit for the search
- **Hypotheses:** "For a feature that predicts S&P 500 returns in 2001-2015, give competing explanations (risk premium, reversal, microstructure, data mining) and what each predicts for 2016-2026 and for other indices."
- **Data and code:** "Write Python that builds 34 technical features from data/market/GSPC.INDX.csv and VIX.INDX.csv with information up to day t only, and reports Newey-West t -statistics for 1-, 5- and 20-day returns in 2001-2015."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); first reproduce the case study: "always up" 58.1%

Appendix: The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance: "Write the pre-analysis plan: the full list of features and horizons, the holdout, the corrections (Holm, BH, max-t) and the result that would count as a failure."
 - ▶ Placebo: shifting the target circularly (250–3,000 days) breaks its alignment with the features but keeps each series' own dependence; a valid null for $\max_j |t_j|$ if the target is stationary and the shifts exceed its dependence range
- **Critique**: "Act as a referee: which of these significant features are one effect counted several times, and which are known anomalies rather than new ones?"
- **Report**: all M tests, all trials including the discarded ones, DSR and PBO for any strategy built on them

References (1/9)

- ▣ Ané, T., Geman, H. (2000). Order flow, transaction clock, and normality of asset returns. *Journal of Finance*, 55(5), 2259–2284.
- ▣ Ansari, A.F. et al. (2024). Chronos: learning the language of time series. *arXiv:2403.07815*.
- ▣ Avramov, D., Cheng, S., Metzker, L. (2023). Machine learning vs. economic restrictions: evidence from stock return predictability. *Management Science*, 69(5), 2587–2619.
- ▣ Bailey, D.H., Borwein, J., López de Prado, M., Zhu, Q.J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4), 39–69.
- ▣ Bailey, D.H., López de Prado, M. (2012). The Sharpe ratio efficient frontier. *Journal of Risk*, 15(2), 3–44.
- ▣ Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ Będowska-Sójka, B., Wójcik, P., Pele, D.T. (2026). Early warning systems for cryptocurrency markets: predicting ‘zombie’ assets using machine learning. *North American Journal of Economics and Finance*, 81, 102543.

References (2/9)

- ▣ Belloni, A., Chernozhukov, V., Hansen, C. (2014). [Inference on treatment effects after selection among high-dimensional controls](#). *Review of Economic Studies*, 81(2), 608–650.
- ▣ Benjamini, Y., Hochberg, Y. (1995). [Controlling the false discovery rate: a practical and powerful approach to multiple testing](#). *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Bergmeir, C., Hyndman, R.J., Koo, B. (2018). [A note on the validity of cross-validation for evaluating autoregressive time series prediction](#). *Computational Statistics & Data Analysis*, 120, 70–83.
- ▣ Breiman, L. (2001). [Random forests](#). *Machine Learning*, 45(1), 5–32.
- ▣ Buehler, H., Gonon, L., Teichmann, J., Wood, B. (2019). [Deep hedging](#). *Quantitative Finance*, 19(8), 1271–1291.
- ▣ Campbell, J.Y., Thompson, S.B. (2008). [Predicting excess stock returns out of sample: can anything beat the historical average?](#) *Review of Financial Studies*, 21(4), 1509–1531.
- ▣ Campbell, M., Hoane, A.J., Hsu, F. (2002). [Deep Blue](#). *Artificial Intelligence*, 134(1–2), 57–83.

References (3/9)

- ▣ Chen, A.Y., Lopez-Lira, A., Zimmermann, T. (2022). Does peer-reviewed research help predict stock returns? *arXiv:2212.10317*.
- ▣ Chen, L., Pelger, M., Zhu, J. (2024). Deep learning in asset pricing. *Management Science*, 70(2), 714–750.
- ▣ Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrics Journal*, 21(1), C1–C68.
- ▣ Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- ▣ Clark, T.E., West, K.D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.
- ▣ Das, A., Kong, W., Sen, R., Zhou, Y. (2024). A decoder-only foundation model for time-series forecasting. *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235.
- ▣ DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3), 837–845.

References (4/9)

- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Dixon, M.F., Halperin, I., Bilokon, P. (2020). *Machine Learning in Finance: From Theory to Practice*. Springer.
- ▣ Fama, E.F., French, K.R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- ▣ Feng, G., Giglio, S., Xiu, D. (2020). Taming the factor zoo: a test of new factors. *Journal of Finance*, 75(3), 1327–1370.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Friedman, J.H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.

References (5/9)

- Hansen, L.P., Jagannathan, R. (1997). Assessing specification errors in stochastic discount factor models. *Journal of Finance*, 52(2), 557–590.
- Hansen, P.R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4), 365–380.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Hastie, T., Tibshirani, R., Friedman, J. (2009). *The Elements of Statistical Learning*. 2nd ed. Springer.
- Hoerl, A.E., Kennard, R.W. (1970). Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67.
- Hooker, G., Mentch, L., Zhou, S. (2021). Unrestricted permutation forces extrapolation: variable importance requires at least one more model, or there is no free variable importance. *Statistics and Computing*, 31(6), 82.
- Hosking, J.R.M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176.
- Kelly, B., Malamud, S., Zhou, K. (2024). The virtue of complexity in return prediction. *Journal of Finance*, 79(1), 459–503.

References (6/9)

- Kelly, B., Xiu, D. (2023). *Financial machine learning*. *Foundations and Trends in Finance*, 13(3–4), 205–363.
- Künsch, H.R. (1989). *The jackknife and the bootstrap for general stationary observations*. *Annals of Statistics*, 17(3), 1217–1241.
- Ledoit, O., Wolf, M. (2008). *Robust performance hypothesis testing with the Sharpe ratio*. *Journal of Empirical Finance*, 15(5), 850–859.
- Lo, A.W. (2002). *The statistics of Sharpe ratios*. *Financial Analysts Journal*, 58(4), 36–52.
- Lo, A.W. (2004). *The adaptive markets hypothesis*. *Journal of Portfolio Management*, 30(5), 15–29.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- López de Prado, M. (2020). *Machine Learning for Asset Managers*. Cambridge University Press.

References (7/9)

- ▣ Lundberg, S.M., Lee, S.-I. (2017). [A unified approach to interpreting model predictions](#). *Advances in Neural Information Processing Systems*, 30.
- ▣ McCracken, M.W., Ng, S. (2016). [FRED-MD: a monthly database for macroeconomic research](#). *Journal of Business & Economic Statistics*, 34(4), 574–589.
- ▣ Nagel, S. (2025). [Seemingly virtuous complexity in return prediction](#). NBER Working Paper 34104.
- ▣ Newey, W.K., West, K.D. (1987). [A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix](#). *Econometrica*, 55(3), 703–708.
- ▣ Novy-Marx, R., Velikov, M. (2025). [AI-powered \(finance\) scholarship](#). NBER Working Paper 33363.
- ▣ Opdyke, J.D. (2007). [Comparing Sharpe ratios: so where are the p-values?](#) *Journal of Asset Management*, 8(5), 308–336.
- ▣ Pesaran, M.H., Timmermann, A. (1992). [A simple nonparametric test of predictive performance](#). *Journal of Business & Economic Statistics*, 10(4), 461–465.
- ▣ Politis, D.N., Romano, J.P. (1994). [The stationary bootstrap](#). *Journal of the American Statistical Association*, 89(428), 1303–1313.

References (8/9)

- ▣ Rasul, K. et al. (2023). [Lag-Llama: towards foundation models for probabilistic time series forecasting.](#) *arXiv:2310.08278*.
- ▣ Robinson, P.M. (1995). [Gaussian semiparametric estimation of long range dependence.](#) *Annals of Statistics*, 23(5), 1630–1661.
- ▣ Romano, J.P., Wolf, M. (2005). [Stepwise multiple testing as formalized data snooping.](#) *Econometrica*, 73(4), 1237–1282.
- ▣ Rosenblatt, F. (1958). [The perceptron: a probabilistic model for information storage and organization in the brain.](#) *Psychological Review*, 65(6), 386–408.
- ▣ Shimotsu, K. (2010). [Exact local Whittle estimation of fractional integration with unknown mean and time trend.](#) *Econometric Theory*, 26(2), 501–540.
- ▣ Shimotsu, K., Phillips, P.C.B. (2005). [Exact local Whittle estimation of fractional integration.](#) *Annals of Statistics*, 33(4), 1890–1933.
- ▣ Tibshirani, R. (1996). [Regression shrinkage and selection via the lasso.](#) *Journal of the Royal Statistical Society B*, 58(1), 267–288.

References (9/9)

- ▣ Wager, S., Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523), 1228–1242.
- ▣ Welch, I., Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.
- ▣ White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126.
- ▣ Woo, G. et al. (2024). Unified training of universal time series forecasting transformers. *arXiv:2402.02592*.
- ▣ Zou, H., Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society B*, 67(2), 301–320.
- ▣ **Code and data:** github.com/danpele/MFM/tree/main/Quantlets/Ch_13